铣刀的切削参数(1)
Explanation on Milling Conditions (1)

切削速度(Vc) Cutting Speed (Vc)

切削速度会因铣刀的材质、刃径、刃长、有效长、加工材料、使用设备、刀柄刚性、加工形状、精度、切削油等不同而存在差异，大致可参考铣刀的材质和加工材料的种类如表1所示。

Appropriate Cutting Speed should be decided by parameters such as tool material, diameter, length of cut, work material,cutting machine, rigidity of tool holder, machining configuration, accuracy, cutting fluid, and etC.Generally tool material and work material are main factors to determine the Cutting Speed.

表 1 切削速度
Table 1. Cutting Speed

加工材料 Work Materials	切削速度[m/min] Cutting Speed (m/min)	
	钨钢铣刀 Solid Carbide Tool	涂层钨钢铣刀 Coated Carbide Tool
碳素钢(S50C等) Carbon Steels	30~60	60~100
合金钢(SCM, SKD等) Alloy Steels	30~40	60~100
调质钢(NAK, HPM等) Prehardened Steels	30~40	60~100
不锈钢(SUS304等) Stainless Steels	20~30	40~80
淬火钢 (SKD61, STAVAX等45~60HRC) Hardened Steels	–	20~100

每刃进给量 fz(mm/Tooth) Feed per Tooth (fz)

进给速度在考虑加工效率时是重要的因素之一，虽然决定其进给速度的每刃进给量因铣刀的刃径、形状、加工材料、使用设备、刀柄刚性、加工形状、精度、切深量不同而存在差异，短刃型铣刀的刃径和刀刃数的大致参考标准如表2所示。

Feed per Tooth is an important element for efficient machining which should be determined by parameters such as tooldiameter, type, work material, cutting machine, rigidity of tool holder, machining configuration, accuracy and cutting depth.Table 2 is a guideline of Feed per Tooth for short flute end mills.

表2每刃进给量
Table 2. Feed per Tooth

刃径[mm] Dia. (mm)	每刃进给量[mm/tooth] Feed per Tooth(mm/tooth)	
	2刃 2-Flutes	4刃 4-Flutes
1	0.001 ~ 0.005	
6	0.02 ~ 0.04	0.01 ~ 0.03
10	0.04 ~ 0.08	0.03 ~ 0.06
20	0.08~0.12	0.06~ 0.1

参考上述的切削速度和每刃进给量，考虑切削相关的所有因素-加工形状、 精度、使用设备、刀柄刚性等,来决定主轴转速和进给速度。

Referring above parameters of Cutting Speed and Feed per Tooth, both Spindle Speed and Feed are calculated considering all other related factors as well.

铣刀的切削参数(2)
Explanation on Milling Conditions (2)

切屑排出量 Q (cm^3 /min) Metal RemovalRate

铣床的单位时间（ 1 分钟 ）内所生成的切屑排出量按照以下公式进行计算。
Metal removal rate (the volume of metal removed in cubic mm per minute) is calculated by the following formula.

$$Q = \frac{ap \times ae \times Vf}{1,000}$$

Q :切屑排出量(cm3/min)
ap :步距量(mm)
Q : Metal removal rate (cm3/min)
ae : Radial depth of cut (mm)

ae :轴向切入量(mm)
vf :进给速度(mm/min)
ap : Axial depth of cut (mm)
Vf : Feed (mm/min)

基于球头铣刀的球头半径(R)和步距(ae)的尖点高度(理论表面粗糙度) all end mill and pick feed (ae) (theoretical surface finish)

球头铣刀加工中，如下图所示，依球头半径和步距而形成的切削剩余部分的高度被称为尖点高度或者残留高度，该高度可以作为理论，上的表面粗糙度进行考量。进行平坦部分的表面加工时,由于步距与步距量ae相同，按照以下的公式进行计算。

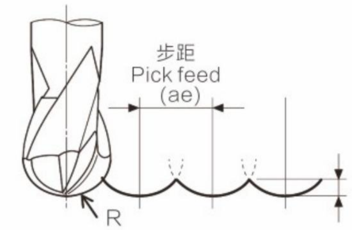
In ball end mill milling, height h of the unmachined part like mountain shape below be formed by ball radius and pick feed,which are called cusp height or scallop height and which is considered theoretical surface finish. Since pick feed is the same ae radial depth of cut at the milling in flat part, it is calculated by the following formula.

$$h=\left(R-\sqrt{R^2-\left(\frac{ae}{2}\right)^2}\right) \times 1000 \fallingdotseq ae^2 \times 1000 \div 8R$$

h:尖点高度(μm)
h: Cusp height (μim)

R:球头半径(mm)
R : Radius (mm)

ae :步距量(mm)
ae : Radial depth of cut (mm)



尖点高度-览表 Cusp Height Quick Reference Matrix

单位Unit : μm

球头半径R Radius	步距量ae[mm] Radial depth of cut ae [mm]										
	0.005	0.01	0.02	0.03	0.04	0.05	0.08	0.01	0.15	0.2	0.3
0.05	0.06	0.25	1.01	2.30	–	–	–	–	–	–	–
0.1	0.03	0.13	0.50	1.13	2.02	3.18	–	–	–	–	–
0.3	0.01	0.04	0.17	0.38	0.67	1.04	2.68	4.20	–	–	–
0.5	0.01以下	0.03	0.10	0.23	0.40	0.63	1.60	2.51	5.66	–	–
0.75	–	0.02	0.07	0.15	0.27	0.42	1.07	1.67	3.76	6.70	–
1	–	0.01	0.05	0.11	0.20	0.31	0.80	1.25	2.82	5.01	–
1.5	–	0.01以下	0.03	0.08	0.13	0.21	0.53	0.83	1.88	3.34	7.52
2	–	–	0.03	0.06	0.10	0.16	0.40	0.63	1.41	2.50	5.63
2.5	–	–	0.02	0.05	0.08	0.13	0.32	0.50	1.13	2.00	4.50
3	–	–	0.02	0.04	0.07	0.10	0.27	0.42	0.94	1.67	3.75

※上表是使用球头铣刀对平坦部进行精加工时根据步距量ae求得的理论值。
※In finishing in flat part, the above values in the table are theoretical values by the radial depth of cut (ae).

刃数选择标准 Selection of Number of Flute

	2刃 2-Flutes	3刃 3-Flutes	4刃 4-Flutes	6刃 6-Flutes
侧面加工 Side Milling	◎	◉	○	×
沟槽加工 Slotting	○	◎	◉	◉

通常, 2刃及3刃排屑槽比较大适合用于沟槽加工。
侧面加工时，因无需担心切屑的堆积，使用4刃及6刃等多刀刃的刀具效果更佳。
Generally 2-flutes and 3-flutes are selected for slotting because of the larger chip pocket.
4-flutes and 6-flutes are recommended for side milling as no problem of chip disposal.

螺旋角选择标准 Selection of Helix Angle

		30°螺旋角 Helix 30°	35°螺旋角 Helix 35°	45°螺旋角 Helix 45°
锋利度Shearing ability		○	◎	◎
抑制振刀效果Chatter resistance		○	◎	◎
表面粗糙度Surface roughness		○	◎	◎
加工面倾斜量Inclination		◎	◎	○
加工面起伏量Wave		◎	◎	○
用途 Application	沟槽加工Slotting	◎	○	△
	侧面加工Side milling	◎	◎	◉
	淬火材料Hardened stees	△	○	◉